

MANGALORE UNIVERSITY



State Education Policy–2024
[SEP-2024]

CURRICULUM STRUCTURE

FOR

BCA

BACHELOR OF COMPUTER APPLICATIONS

MANGALOREUNIVERSITY

Suggested programme structure for the Under Graduate Programmes

[BCA, BCA(A.I&M.L), BCA(D.A)]

Semester	Course1	Course2	Course3	Elective/Optional	Course	Language	Compulsory	Total Credit	Total Working hour
I	5 (3T+2P)	5 (3T+2P)	5 T			3+3	2	23	4+4+4+4+5+4+4+2=31
II	5 (3T+2P)	5 (3T+2P)	5T			3+3	2	23	4+4+4+4+5+4+4+2=31
III	5 (3T+2P)	5 (3T+2P)	5T	2		3+3		23	4+4+4+4+5+4+4+2=31
IV	5 (3T+2P)	5 (3T+2P)	5T	2		3+3	2	25	4+4+4+4+5+2+4+4+2=33
V	8[(2x3T)+2P]]	8[(2x3T)+2P]]	8[(2x3T)+2P]				2	26	3+3+4+3+3+4+3+3+4+2=32
VI	3T	3T	3T		3T	Projectwork 12		24	3+3+3+3+24=36
								144	

Note:

- Course1 and Course2: I to IV Semester: Theory 3 credit=4 contact hours & Practical 2 credit=4 contact hours
- Course3: I to IV Semester: Theory 5 credit=5 contact hours
- Course1, Course2 and Course3: V and VI Semester: Theory 3 credit=3 contact hours & Practical 2 credit=4 contact hours
- Elective/Optional: 2 credit=2 contact hours
- Languages: 3 credit=4 contact hours
- Compulsory: 2 credit=2 contact hours

CURRICULUM STRUCTURE FOR IT O VI SEMETER BCA

Semester I								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1		Fundamentals of Computers	Theory	4	80	20	100	3
2		Information Technology Lab	Practical	4	40	10	50	2
3		Programming in C	Theory	4	80	20	100	3
4		C Programming Lab	Practical	4	40	10	50	2
5		Discrete Mathematics for Computer Applications	Theory	5	80	20	100	5
6		Language-I	Lang	4	80	20	100	3
7		Language-II	Lang	4	80	20	100	3
8		Constitutional Values	Compulsory	3	40	10	50	2
Sub- Total				32	520	130	650	23
Semester II								
9		Data Structures	Theory	4	80	20	100	3
10		Data Structures Lab	Practical	4	40	10	50	2
11		Object Oriented Programming using Java	Theory	4	80	20	100	3
12		Object Oriented Programming Lab	Practical	4	40	10	50	2

13		Computational Mathematics	Theory	5	80	20	100	5
14		Language-I	Lang	4	80	20	100	3
15		Language-II	Lang	4	80	20	100	3
16		Environmental studies	Compulsory	3	40	10	50	2
Sub- Total				32	520	130	650	23

Semester III								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1		Language-I	Lang	4	80	20	100	3
2		Language-II	Lang	4	80	20	100	3
3	BCA —3.1	Database Management System	Core	4	80	20	100	3
4	BCA -3.2	C#andDotnet Framework	Core	4	80	20	100	3
5	BCA-3.3	Computer Networks	Core	5	80	20	100	5
6	BCA-3.4	Database Management System Lab	practical	4	40	10	50	2
7	BCA-3.5	C# and Dotnet Framework Lab	practical	4	40	10	50	2
8	BCA-3.6	A) OpenSource Tools B) WebContent Management System C) DEVOPS	Elective	2	40	10	50	2
Sub- Total				31	520	130	650	23

Semester IV								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1		Language-I	Lang	4	80	20	100	3
2		Language-II	Lang	4	80	20	100	3
3	BCA –4.1	Python Programming	Core	4	80	20	100	3
4	BCA –4.2	Advanced JAVA and J2EE	Core	4	80	20	100	3
5	BCA –4.3	Operating System Concepts	Core	5	80	20	100	5
6	BCA –4.4	Python Programming Lab	practical	4	40	10	50	2
7	BCA –4.5	Advanced JAVA and J2EE Lab	practical	4	40	10	50	2
8	BCA –4.6	A) Distributed Computing B) Object Oriented Analysis & Design C) Digital Image Processing	Elective	2	40	10	50	2
9	BCA –4.7	Internet Basics	Compulsory	2	40	10	50	2
Sub- Total				33	560	140	700	25

Semester V								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BCA-5.1	Software Engineering	Core	3	80	20	100	3
2	BCA-5.2	Web Development	Core	3	80	20	100	3
3	BCA-5.3	Design Analysis and Algorithm	Core	3	80	20	100	3
4	BCA-5.4	Statistical Computing and R Programming	Core	3	80	20	100	3
5	BCA-5.5	Cloud Computing	Core	3	80	20	100	3
6	BCA-5.6	Cryptography and Network Security	Core	3	80	20	100	3
7	BCA-5.7	Design Analysis and Algorithm Lab	Practical	4	40	10	50	2
8	BCA-5.8	R Programming Lab	Practical	4	40	10	50	2
9	BCA-5.9	Web Development Lab	Practical	4	40	10	50	2
10	BCA-5.10	Data Analytics using Excel	Compulsory	2	40	10	50	2
Sub- Total				32	640	160	800	26

Semester VI								
Sl. No	Course Code	Title of the Course	Category of Courses	Teaching Hours per Week	SEE	IA	Total Marks	Credits
1	BCA- 6.1	Artificial Intelligence and Applications	Core	3	80	20	100	3
2	BCA- 6.2	Principles Cyber Security	Core	3	80	20	100	3
3	BCA- 6.3	Fundamentals of Data Science	Core	3	80	20	100	3
4	BCA- 6.4	Digital Marketing	Core	3	80	20	100	3
4	BCA- 6.5	Project Work	Project Work	24	300	100	400	12
Sub- Total				36	620	180	800	24

SEMESTER-I

ProgramName	BCA	Semester	I
CourseTitle	Fundamentals of Computers (Theory)		
CourseCode:	BCA-1.1	No. of Credits	03
Contact hours	4 Hours per week	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After completing this course satisfactorily, a student will be able to:

- Understand the fundamentals of computer system
- Identify different components within the computer system
- Understand different types of input and output devices
- Demonstrate the working concepts of different devices connected to computer
- Explain different generations of programming languages and their significance
- Understand the use of Word processing, Spreadsheet, Presentation and DBMS applications
- Understand Digital computer and digital systems functioning

Unit	Description	Hours
1	Computer Basics: Introduction, Characteristics computers, Evolution computers, Generations of computers, Classification of computers, the computer system, Application of computers. Computer Architecture: Introduction, Central processing unit- ALU, Registers, Control unit, system bus, main memory unit, cache memory Input devices: Introduction, Types of input devices, Keyboard, Mouse, Track ball, Joystick light pen, Touch screen and track pad. Speech recognition, digital camera, webcam, flatbed scanner Output devices: Types of output, Classification of output	13

	devices, Printers–Dotmatrix, Ink-jet, Laser, Hydra, Plotter, Monitor – CRT, LCD, Differences between LCD and CRT	
2	Computer software: Introduction, software definition, relationship between software and hardware, software categories Computer programming languages: Introduction, developing a program, Program development cycle, Types of programming languages, generation of programming languages, Features of a good programming language. Problem Solving techniques: Introduction, Problem solving procedure. Algorithm: Steps involved in algorithm development, Algorithms for simple problems (To find largest of three numbers, factorial of a number, check for prime number, check for palindrome, Count number of odd, even and zeros in a list of integers) Flowcharts: Definition, advantages, Symbols used in flow charts. Flowcharts for simple problems mentioned in algorithms. Pseudocode.	13
3	Digital Computers and Digital System: Introduction to Number System, Decimal number, Binary number, Octal and Hexadecimal numbers, Number base conversion, Complements, Binary codes, Binary arithmetic, Addition, Subtraction in the 1's and 2's complement system, Subtraction in the 9's and 10's complement system. Boolean Algebra: Basic definitions, Axiomatic definition of Boolean algebra, Basic theorems and properties of Boolean algebra, Venn diagram.	13
4	Digital logical gate: Boolean functions, Canonical and Standard forms, Minterms, Maxterms, other logic operations, Digital logic gates, Universal gates. Simplification of Boolean function: The map method, Two and three variable maps, Four variable maps, Don't care conditions, Product of sum simplification.	13
Text Books: 1. ITL Education Solution Limited, Introduction to Information Technology, Second Edition, Pearson 2. M. Morris Mano, Digital Logic and Computer design, PHI, 2015		

ReferencesBooks:

1. PradeepK.SinhaandPritiSinha,ComputerFundamentals,SixthEdition,BPB Publication.
2. DavidRileyandKennyHunt,Computationalthinkingformodernsolver,Chapman & Hall/CRC.
3. J. Glenn Brookshear, Computer Science: An Overview, Twelfth Edition, Addison-Wesley
4. R.G.Dromey, Howto solveitbyComputer,PHI.

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Role Plays/ Think-Pair-Share/ Predict-Observe- Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/ Jigsaw/ Field based Learning/ Project Based Learning/Mini Projects/ Hobby Projects/ Forum Theatre/ Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA	Semester	I
CourseTitle	Programming in C (Theory)		
CourseCode:	BCA-1.2	No. of Credits	03
Contact hours	4 Hours per week	Duration of SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

Course Outcomes (COs):

After completing this course satisfactorily, a student will be able to:

- Confidently operate Desktop Computers to carry out computational tasks
- Understand working of Hardware and Software and the importance of operating systems
- Understand programming languages, number systems, peripheral devices, networking, multimedia and internet concepts
- Read, understand and trace the execution of programs written in C language
- Write the C code for a given problem
- Perform input and output operations using programs in C
- Write programs that perform operations on arrays

Unit	Description	Hours
1	Overview of C: History of C, Importance of C Program, Basic structure of a C-program, Execution of C Program. C Programming Basic Concepts: Character set, C token, Keywords and identifiers, Constants, Variables, data types, Declaration of variables, assigning values to variables, defining symbolic constants. Input and output with C: Formatted I/O functions - printf and scanf, control strings and escape sequences, output specifications with printf functions; Unformatted I/O functions to read and display single character and a string - getchar, putchar, gets and puts functions.	13

2	Operators & Expressions: Arithmetic operators; Relational operators; Logical operators; Assignment operators; Increment & Decrement operators; Bitwise operators; Conditional operator; Special operators; Operator Precedence and Associativity; Evaluation of arithmetic expressions; Type conversion. Control Structures: Decision Making and Branching - Decision making with if statement, simple if statement, the if else statement, nesting of if ... else statements, the else if ladder, the switch statement, the ?: operator, the go to statement. Decision making and looping - The while statement, the do statement, for statement, nested loops, exit, break, jumps in loops.	13
3	Derived data types in C: Arrays - declaration, initialization and access of one-dimensional and two-dimensional arrays. programs using one- and two-dimensional arrays, sorting and searching arrays. Handling of Strings: Declaring and initializing string variables, reading strings from terminal, writing strings to screen, Arithmetic operations on characters, String handling functions - strlen, strcmp, strcpy, strstr and strcat; Character handling functions - toascii, toupper, tolower, isalpha, isnumeric etc. User-defined functions: Need for user-defined functions, Declaring, defining and calling C functions, return values and their types, Categories of functions: With/without arguments, with/without return values. Nesting of functions. Recursion: Definition, example programs. Storage classes: Automatic, Global, Static, Register.	13
4	Pointers: Understanding pointers, accessing the address of a variable, declaring and initializing pointers, accessing a variable through its pointer, pointer expression, pointer increments and scale factor, pointers and arrays, pointer and strings. Structures and unions: Structure definition, giving values to members, structure initialization, comparison of structure variables, arrays of structures, arrays within structures, Structure and functions, structures within structures. Unions File Handling in C: Create in Read/Write and Append mode, copying file.	13

	ThePre-processor: Macrosubstitution,fileinclusion.	
TextBook: 1. E.Balagurusamy,ProgramminginANSIC,7thEdition,TataMcGrawHill Reference Books: 1. HerbertSchildt,C:TheCompleteReference,4thEdition 2. BrainW.Kernighan,CProgrammingLanguage,2ndEdition,PrenticeHall Software 3. Kernighan&Ritchie:TheCProgrammingLanguage,2nd Edition,PHI 4. Kamthane,ProgrammingwithANSIandTURBOC,PearsonEducation 5. V.Rajaraman,ComputerProgramminginC,2ndEdition,PHI 6. S.ByronGottfried,ProgrammingwithC,2ndEdition,TMH 7. YashwantKanitkar,LetusC,15th Edition,BPB 8. P.B.Kottur,ComputerConceptsandProgramminginC,23rdEdition,Sapna BookHouse		

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Role Plays/ Think-Pair-Share/ Predict-Observe- Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/ Jigsaw/ Field based Learning/ Project Based Learning/Mini Projects/ Hobby Projects/ Forum Theatre/ Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA	Semester	I
CourseTitle	DiscreteMathematicsforComputerApplications(Theory)		
CourseCode:	BCA-1.3	No.ofCredits	05
Contacthours	5Hours perweek	Durationof SEA/Exam	3 Hours
Formative AssessmentMarks	20	Summative AssessmentMarks	80

CourseOutcomes (COs):

Afterthesuccessfulcompletionofthecourse,the studentwillbeableto:

- Study and solve problems related to connectives, predicates and quantifiers under different situations.
- UnderstandthebasicconceptsofDiscreteProbability.
- Todeveloptheknowledgeaboutderivativesandknowvariousapplicationsof differentiation.
- UnderstandtheApplicationsofDiscreteMathematicsinModelling Computation.
- UnderstandthebasicconceptsofMathematicalreasoning,setandfunctions

Unit	Description	Hours
1	Mathematical logic: Introduction, statements, Connectives, negation, conjunction, disjunction, statement formulas and truth tables, conditional and bi Conditional statements, tautology, contradiction, equivalence of formulas, dualitylaw, Predicates and Quantifiers, arguments, joint Daniel Sets: Definition, notation, inclusion and equality of sets, the power set, Operations on sets,Venn diagram, ordered pairs, and n-tuples, Cartesian product,	15

	Relations: Introduction, properties of a binary relation in a set, Relation matrix and graph of a relation, equivalence relations, compatibility relations, composition of Binary relation	
2	Partial Ordering: Definition, lexicographic ordering, Partially ordered set, Hasse diagram, well-ordered set Functions: Definition and introduction, types of functions, composition of functions, inverse functions Counting: Basics of counting, Pigeonhole principle, Permutation and combination, Generalized Permutations and Combinations, generating permutation and combination, inclusion and exclusion	15
3	Discrete Probability: Introduction, finite probability, probabilities of complements and unions of events, probability theory, conditional probability, independence, random variables, Bayes' theorem, expected value and variance, independent random variable. Mathematical Induction: Mathematical Induction, principle of mathematical induction, proving inequalities, strong induction and well ordering Number Theory: Division algorithm, Modular arithmetic, primes and greatest common divisors, least common multiple, the Euclidean algorithm	15
4	Graphs: Graphs and Graph models, Graph Terminology and Special Types of Graphs, Representing Graphs and Graph Isomorphism, Connectivity, Euler and Hamilton Paths, Shortest-Path Problems, Planar Graphs, Graph Coloring. Trees: Directed tree, leaf node, branch node, ordered tree, degree of a node, forest, descendent, m-ary tree, conversion of directed tree into a binary tree. Applications of Discrete Mathematics in Modelling Computation: Language and Grammars – Introduction, Phrase-Structured, Types, Derivation Trees; Finite State Machines with Output – Introduction, Finite State Machines, Types; Finite State Machines without Output - Introduction, Set of Strings, Finite State Automata, Language Recognition by FSM; Language Recognition – Introduction; Turing Machine – Introduction, Definition	15

TextBooks:

1. J.P. Trembley and R. Manobar, Discrete Mathematical Structures, McGraw Hill Education Private Limited, New Delhi.
2. Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, 2012.
3. Bernard Kolman, Robert C. Busby, Sharon Ross, Discrete Mathematical Structure, 2003.
4. C. L. Liu, D. P. Mohapatra, Elements of Discrete Mathematics, 4th Edition McGraw Hill Education Private Limited, New Delhi.

Reference Books:

1. D C Sanchethi and V K Kapoor, Business Mathematics, Eleventh Revised Edition, Sulthan Chand & Sons Educational publishers, New Delhi,
2. Narsingh Deo, Graph Theory with Applications to Engg and Comp. Sci, PHI, 1986.
3. Ralph P. Grimaldi, B. V. Ramatta, Discrete and Combinatorial Mathematics, 5th Edition, Pearson, Education
4. K Chandrashekhar Rao, Discrete Mathematics, Narosa Publishing House, New Delhi

**Ped
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gy: Lecture/ PPT/ Videos/ Animations/ Role Plays/ Think-Pair-Share/ Predict-Observe-Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/ Jigsaw/ Field based Learning/ Project Based Learning/ MiniProjects/ HobbyProjects/ Forum Theatre/Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA-	Semester	I
CourseTitle	InformationTechnologyLab		
CourseCode:	BCA-1.4	No.ofCredits	02
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative AssessmentMarks	10	Summative AssessmentMarks	40

PART-A:MSWORD

1. Prepareadocumentusingdifferentformattingtools

Highlights of the National Education Policy (NEP) 2020



Note4Students

From UPSC perspective, the following things are important :

Prelims level : National Education Policy

Mains level : Need for imbibing competitiveness in Indian education system

New Policy aims for **universalization of education** from pre-school to secondary level with 100 % Gross Enrolment Ratio (GER) in school education by 2030. NEP 2020 will bring 2 crores out of school children back into the mainstream through the open schooling system.

- ❖ The current 10+2 system to be replaced by a **new 5+3+3+4 curricular structure** corresponding to ages 3-8, 8-11, 11-14, and 14-18 years respectively. **This will bring the hitherto uncovered age group of 3-6 years under the school curriculum, which has been recognized globally as the crucial stage for the development of mental faculties of a child.**
- ❖ The new system will have 12 years of schooling with three years of Anganwadi/ pre-schooling.
 - o Emphasis on Foundational Literacy and Numeracy, no rigid separation between academic streams, extracurricular, vocational streams in schools; Vocational Education to start from Class 6 with Internships
 - o Teaching up to at least Grade 5 to be in mother tongue/ regional language. No language will be imposed on any student.
- Assessment reforms with **360° Holistic Progress Card**, tracking Student Progress for achieving Learning Outcomes
- A new and comprehensive National Curriculum Framework for Teacher Education, NCFTE 2021, will be formulated by the NCTE in consultation with NCERT.
- By 2030, the minimum degree qualification for teaching will be a 4-year integrated B.Ed. degree.
- Gross Enrolment Ratio in higher education to be raised to **50% by 2035; 3.5 crore seats to be added in higher education.**
- The policy envisages broad-based, multi-disciplinary, holistic Under Graduate Program with flexible curricula, creative combinations of subjects, integration of vocational education and multiple entries and exit points with appropriate certification.
- **Academic Bank of Credits to be established to facilitate Transfer of Credits**

Multidisciplinary Education and Research Universities (MERUs), at par with IITs, IIMs, to be set up as models of best multidisciplinary education of global standards in the country.

Affiliation of colleges is to be **phased out in 15 years** and a stage-wise mechanism is to

be established for granting graded autonomy to colleges.

Over a period of time, it is envisaged that every college would develop into either an Autonomous degree-granting College or a constituent college of a university.

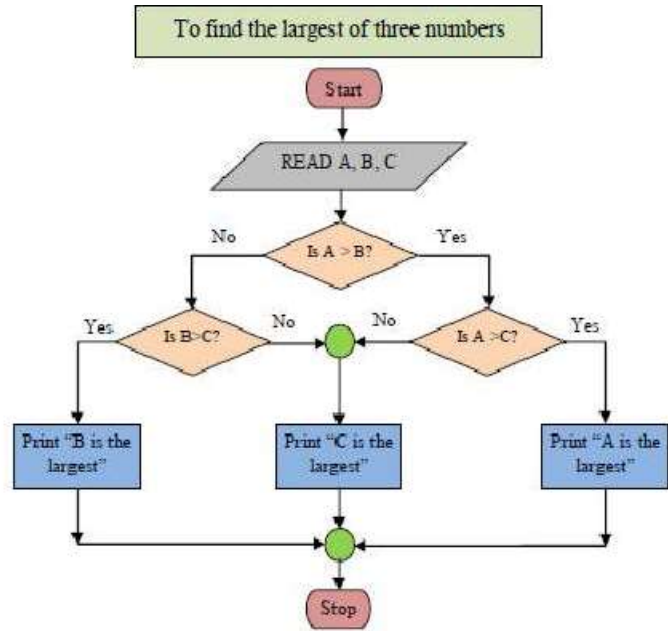
$$\frac{df}{dt} = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

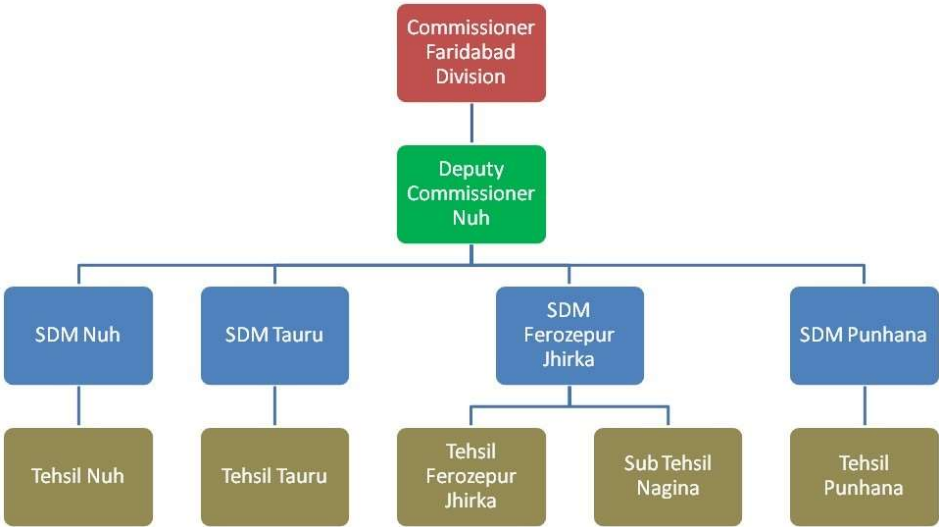
$$(a-b)^2 = (a+b)^2 - 4ab$$

$$a^2 + b^2 = (a-b)^2 + 2ab$$

2. Prepare a document using SmartArt and Shape tools



Organization Chart—Administration Faridabad Division



3. Prepare a document with table to store sales details of a company for different quarters and calculate total, average and find maximum, minimum sales value.

Branch Code	Branch	Sales in Quarters				Total	Avg
		1	2	3	4		
A101	Mangalore	354690	244610	383290	413670		
A102	Udupi						
Total (Across Branches)							
Average (Across Branches)							
Highest Sales (Across Branches)							
Lowest Sales (Across Branches)							

TIMETABLE

Class : I BCA				Room No. 206			
Day	I	II	III	IV		V	VI
Monday					LUNCH BREAK		
Tuesday							
Wednesday							
Thursday							
Friday							
Saturday						***	

4. Prepare interview call letters for five candidates describing about the company and instructions about the interview. Use Mail merge feature

Interview call Letter Format

Date:

[Name of the candidate]

[Address]

Dear [name of the candidate]

This is to the reference of your application for the job [name of the job] indicating interest in seeking employment in our organisation. We thank you for the same.

We would like to inform you that your profile is being shortlisted for the job role and is best suited for it. Therefore, we would like to take a face to face interview with you on [date of interview] at [venue details].

We hope that the venue is suitable for you. If not please get in touch with us, so that we can arrange the date and venue according to your availability.

The company will reimburse you all the expenses incurred by you for this interview. This letter has an attachment in which you need to fill the details and carry it along on the date of interview. Please carry your CV also along with you.

Kindly confirm your availability for the date and venue. If there are any changes to be done, please contact us at phone number: [999xxxx999] and email id: abcnd@mail.com.

We look forward to seeing you.

Regards,

Name of the Manager

Designation Name

Company name

PART-B:MSPOWERPOINT

1. Create a presentation (minimum 5 slides) about your college. It should contain images, chart, Bulleted text... The slides should be displayed automatically in a loop.
2. A simple quiz program. Use hyperlinks to move to another slide in the presentation to display the result and correct answer/wrong answer status. Use at least four questions.

[Navigation must be done by hyperlink]

3. Create a presentation for a business proposal (minimum 5 slides).
 - Slides must include company logo in header
 - A title slide with table of contents
 - financial data of the company in the table
 - Company sales and profit in charts
 - Make use of animation and transition
4. Create a presentation for a college project (minimum 5 slides).
 - Master slide
 - Add comments for each slide
 - Add Audio and video to the slide
 - Add header and footer.
 - Add source citation
 - Make use of animation and transition

[Presentation must include title slide, Module Design, Chart, references]

PART-C:MS EXCEL

(Note: Give proper titles, column headings for the worksheet. Insert 10 records for each exercise in such a way to get the result for all the conditions. Format the numbers appropriately wherever needed).

1. Create a worksheet to maintain student information such as *RollNo, Name, Class, Marks in three subjects* of 10 students. Calculate total marks, average and grade. Find grade for Distinction, First class, Second class, Pass and Fail using normally used conditions.

- Using custom sort, sort the data according to class: - Distinction first, First Class next, and so on. Within each class, average marks should be in descending order.
- Also draw the Column Chart showing the Roll No versus Average scored.

(Note: Worksheet creation and formatting 3 marks, calculations: 3 marks, sorting: 2 marks, chart: 2 marks)

2. Prepare a worksheet to store details of electricity consumed by customers. Details are CustomerNo, CustomerName, MeterNo, Previous meter reading, Current meter reading of 10 customers. Calculate total number of units consumed and total amount to be paid by each consumer using following conditions:

- If unit consumed is up to 30, charge is 100.
- 31 to 100 units, 4.70 per unit
- 101 to 200 units, 6.25 per unit
- Above 200 units, 7.30 per unit.
- Use Data validation to see that current reading is more than previous reading.
- Arrange the records in the alphabetic order of names.
- Filter the records whose bill amount is more than Rs. 1500.

(Note: Worksheet creation and formatting 2 marks, Data validation: 2 marks, calculations: 2 marks, sorting: 2 marks, filtering: 2 marks)

3. Create Employee worksheet having EmpNo, EmpName, DOJ, Department, Designation and Basic Pay of 8 employees. Calculate DA, HRA, Gross Pay, Profession Tax, Net Pay, Provident Fund as per the rule:

- $DA = 30\% \text{ of basic pay}$
- $HRA = 10\% \text{ of basic pay if basic pay is less than } 25000, 15\% \text{ of basic pay otherwise.}$
- $Gross = DA + HRA + Basic\ pay$
- $Provident\ fund = 12\% \text{ of Basic pay or Rs. } 2000, \text{ whichever is less.}$
- $Profession\ Tax = Rs. 100 \text{ if Gross pay is less than } 10000, Rs. 200 \text{ otherwise.}$
- $Net\ Pay = Gross - (Profession\ tax + Provident\ Fund)$

- Using Pivot table, display the number of employees in each department and represent it using Pie chart.

(Note: Worksheet creation and formatting: 2 marks, calculations: 3 marks, Pivot table: 3 marks, Chart: 2 marks)

4. Create a table COMMISSION containing the percentage of commission to be given to salesmen in different zones as follows:

Zone	Percentage
South	10
North	12.5
East	14
West	13

Create another table SALES in the same worksheet to store salesman name, zone name, place, name of the item sold, rate per unit, quantity sold. Calculate total sales amount of each salesman. Referring the COMMISSION table, write the formula to compute the commission to be given. (Hint: Use if function and absolute cell addresses)

Using advanced filtering show the result in other parts of the worksheet.

- Show the records of various zones separately.
- Show the records of only East and West zones.
- Display the details of the items sold more than 50, in South or North zones.

(Note: Worksheet creation and formatting: 2 marks, calculations: 2 marks, filtering: 6 marks)

PART-D: MSACCESS

1. Create Employee database and table Emp using MSACCESS with following Structure

Emp no	Ename	Designation	Deptno	DOJ	Basic Salary
101	RAMESH	MANAGER	10	10/10/2000	25000
102	SMITHA	CLERK	12	12/5/1999	15000
103	DEVIKA	ATTENDER	10	11/9/2001	12000
104	RAJESH	HR	15	15/4/2000	12000
105	GIRISH	SUPERVISOR	12	6/11/2005	18000
106	SATHYA	DRIVER	16	11/9/2001	11000
107	MANOJ	SWEEPER	10	22/6/2006	8000
108	BHOOMIKA	SECURITY	15	12/5/1999	10500
109	KIRAN	CLERK	14	11/9/2001	15000
110	PRATHIKSHA	SUPERVISOR	10	8/8/2005	18000

Perform following operation:

- List all the Employees Who are working in Deptno.10
- List all the Employees who get less than 20000 Salary
- Update Salary by adding the increments as per the following:-
 - 10% Increment in Basic Salary who get <20000
 - 5% Increment in Basic Salary who get >=20000.

2. Create the "Order" database and a table "Orderdtl" having following records:

Order No	Order Date	Order Item	Order Qty	Order Price	Client Code	Delivery Type	Order Status
1011	12/02/2015	LED Monitors	100	750000	1025	Road	Delivered
1012	12/03/2015	CPU	12	500000	1026	SHIP	Not Delivered
1005	15/02/2014	Keyboard	80	48000	1027	Road	Delivered
1010	02/02/2016	LED Monitors	30	64000	1028	Flight	Delivered
1016	19/4/2015	Scanner	40	35000	1029	Road	Delivered
1009	9/05/2018	LED Monitors	25	125000	1030	Flight	Not Delivered
1008	13/8/2017	CPU	25	450000	1031	SHIP	Delivered
1014	1/7/2018	Printer	50	90000	1032	Road	Not Delivered

Execute following Query

- Display all the OrderNo. which have not been yet Delivered.
- Display all the Orders of LED Monitor and CPU.
- Display all the Orders of LED Monitor and CPU which are not have been delivered yet.

3. Create a "Stock" database having "Inventory" table:

Item Code	Item Name	Opening Stock(Qty)	Purchase(Qty)	Sale (Qty)	Closing Stock(Qty)	Remark
101	MONITOR	100	25	35		
102	PRINTER	75	40	15		
103	SCANNER	120	30	20		
104	CPU	50	35	10		
105	KEYBOARD	105	45	55		

Execute following Query

- Calculate the closing stock of each item (Closing Stock = Opening Stock + Purchase – Sales)
- Display all the Items which have closing stock < 100
- If closing stock is less than 100 then set the remark as "Re-Order Level" otherwise "Enough Stock".

4. Create a "Company" database having "Sales" table with fields saleid, quarter, product, no_of_sales.

Perform the followings:

- Design a form to insert records to Sales table
- Generate a report to display Sales details of product based on quarters.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	MSWORD	8 Marks
Program-2	MS POWERPOINT	7 Marks
Program-3	MS EXCEL	10
Program-4	MS ACCESS	10

Practical Record	05Marks
Total	40 Marks

ProgramName	BCA	Semester	I
CourseTitle	CProgrammingLab		
CourseCode:	BCA-1.5	No.ofCredits	02
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

PART-A

1. Program to find the roots of quadratic equation using else if ladder.
2. Program to read two integer values & a operator as character and perform basic arithmetic operations on them using switch case (+, -, *, / operations)
3. Program to reverse a number and find the sum of individual digits. Also check for palindrome.
4. Program to calculate and display the first 'n' Fibonacci numbers
5. Program to find given number is a prime or not.
6. Program to count occurrences of each character in a given string.
7. Program to read string with alphabets, digits and special characters and convert upper case letters to lower case and vice versa and retain the digits and special characters as it is.

8. Program to search for number of occurrences of number in a list of numbers using one-dimensional array also display its positions.

PART-B

1. Program to find the largest and smallest elements with their position in a one-dimensional array.
2. Program to read 'n' integer values into a single dimension array and arrange them in ascending order using bubble sort method.
3. Menu driven Program to perform addition and multiplication of two Matrices
4. Program to find nCr and nPr using recursive function to calculate factorial.
5. Program to read a string and count number of letters, digits, vowels, consonants, spaces and special characters present in it using user defined function
6. Program to sort a list of strings in ascending order using Pointers
7. Program to enter the information of a student like name, register number, marks in three subjects into a structure and display total, average and grade. Display details in a neat form.
8. Write a menu driven program to
 - a. Create a text file
 - b. Append the contents of a text file to another existing file by accepting filenames
 - c. Display the content of entered filename
 - d. Exit

Create two text files during the execution of the program. Display their contents. Perform Appending. Display the contents again. Always check for the existence of the inputted file names.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing: 7 Marks Execution: 8 Marks	15 Marks
Program-2	PART-B Writing: 10 Marks Execution: 10 Marks	20 Marks
Practical Record		05 Marks

Total	40 Marks
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SEMESTER- II

ProgramName	BCA	Semester	II
CourseTitle	Data Structures(Theory)		
CourseCode:	BCA-2.1	No.ofCredits	03
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative AssessmentMarks	20	Summative AssessmentMarks	80

CourseOutcomes (COs):

Afterthesuccessfulcompletionofthecourse,thestudentwillbeableto:

- Describe how arrays, records, linked structures, stacks, queues, trees, and graphsare represented in memory and used by algorithms
- Describecommonapplicationsforarrays,records,linkedstructures,stacks, queues, trees, and graphs
- Write programsthatusearrays,linkedstructures,stacks,queues,trees,andgraphs
- Demonstratedifferentmethodsfortraversingtrees

- Compare alternative implementations of data structures with respect to performance
- Describe the concept of recursion, give examples of its use
- Discuss the computational efficiency of the principal algorithms for sorting, searching, and hashing

Unit	Description	Hours
1	Introduction to data structures: Introduction, Basic terminology; Elementary Data Organization, Data Structures, Data Structure Operations Introduction to Algorithms, Preliminaries: Introduction, Algorithmic notations, Control structure. Recursion: Definition; Recursion Technique Examples – Factorial, Fibonacci sequence, Towers of Hanoi. Arrays: Basic Concepts – Definition, Declaration, Initialisation, Operations on arrays, Types of arrays, Representation of Linear Arrays in memory, Traversing linear arrays, Inserting and deleting elements, Multidimensional arrays- Two Dimensional Arrays Representation of two-dimensional arrays, Sparse matrices. Sorting: Selection sort, Bubble sort, Quick sort, Insertion sort, Merge sort	13
2	Searching: Definition, Sequential Search, Binary search Dynamic memory management: Memory allocation and de-allocation functions - malloc, calloc, realloc and free. Linked list: Basic Concepts – Definition and Representation of linked list, Types of linked lists- Singly linked list, Doubly linked list, Header linked list, Circular linked list, Representation of Linked list in Memory; Operations on Singly linked lists– Traversing, Searching, Insertion, Deletion, Memory allocation, Garbage collection	13
3	Stacks: Basic Concepts – Definition and Representation of stacks- Array representation of stacks, Linked representation of stacks, Operations on stacks, Applications of stacks, Infix, postfix and prefix notations, Conversion from infix to postfix using stack, Evaluation of postfix expression using stack, Application of stack in function calls. Queues: Basic Concepts – Definition and Representation of	13

	queues- Array representation of Queues, Linked representation of Queues, Types of queues - Simple queues, Circular queues, Double ended queues, Priority queues, Operations on queues	
4	Trees: Definition, Tree terminologies—node, root node, parent node, ancestor of a node, siblings, terminal & non-terminal nodes, degree of a node, level, edge, path, depth Binary tree: Type of binary trees—strict binary tree, complete binary tree, binary search tree,; Array representation of binary tree, Traversal of binary tree—preorder, inorder and postorder traversal Graphs: Terminologies, Matrix representation of graphs; Traversal: Breadth First Search and Depth first search.	13
Text Books: 1. Seymour Lipschutz, Data Structures with C, Schaum's Outlines Series, Tata McGraw Hill, 2011 2. R. Venkatesan and S. Lovelyn Rose, Data Structures, First Edition: 2015, Wiley India Pvt. Ltd. Publications Reference Books: 1. Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures, Computer Science Press, 1982. 2. Aaron M. Tenenbaum, Data structures using C, First Edition, Pearson Education 3. Kamathane, Introduction to Data structures, Pearson Education, 2004 4. Y. Kanitkar, Data Structures Using C, Third Edition, BPB 5. Padma Reddy: Data Structure Using C, Revised Edition 2003, Sai Ram Publications. 6. Sudipa Mukherjee, Data Structures using C—1000 Problems and Solutions, McGraw Hill Education, 2007		

Pedagogy:
Lecture/PPT/Videos / Animations/ Role Plays/ Think-

Pair-Share/ Predict-Observe- Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/ Jigsaw/ Field based Learning/ Project Based Learning/ Mini Projects/ Hobby Projects/ Forum Theatre/ Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA	Semester	II
CourseTitle	ObjectOrientedProgrammingusingJava(Theory)		
CourseCode:	BCA-2.2	No.ofCredits	03
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative Assessment Marks	20	Summative Assessment Marks	80

CourseOutcomes (COs):

Afterthesuccessfulcompletionofthecourse,the studentwillbeableto:

- UnderstandthefeaturesofJavaandthearchitectureofJVM
- Write, compile, and execute Java programs that may include basic data types and control flow constructs and how type casting is done
- Identifyclasses, objects, members of a class and relationships among themneeded for a specific problem and demonstrate the concepts of polymorphism and inheritance
- The students will be able to demonstrate programs based on interfaces and threads and explain the benefits of JAVA's Exceptional handling mechanism compared to other Programming Language
- Write, compile, execute Java programs that include GUIs and event driven programming and also programs based on files

Unit	Description	Hours
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1	Fundamentals of Object Oriented Programming: Introduction, Object Oriented Paradigm, Basic Concepts of OOP, Benefits and Applications of OOP. Introduction to Java: Java Features, Java Environment, Simple Java Program, Java Program Structure, Java Tokens, Java Statements, Java Virtual Machine. Java Programming Basics: Constants, Variables, Data Types, Declaration of variables, Giving values to the variable, Scope of variables, Symbolic constants, Type casting. Operators and Expressions: Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operator, Increment and Decrement Operators, Conditional Operator, Special Operators, Mathematical functions. Using I/O: Byte streams and character streams, predefined streams, reading console input, reading characters, strings, writing console output. Decision Making & Branching: Simple if statement, if..else statement, nesting of if..else statement, the else..if ladder, the Switch statement..	13
2	Decision making & Looping- The while statement, the do statement, the for statement . Jumps in loops, Labelled loops. Class & Objects- Class Fundamentals, Declaring Objects, Assigning Object Reference Variables, Introducing Methods, Constructors, The 'this' keyword, Overloading Methods, Using Objects as Parameters, Returning Objects, Recursion, Understanding 'static', Introducing 'final', Using Command-Line Arguments, Varargs : Variable-Length Arguments Arrays and Strings: One dimensional arrays, Creating an array, Two dimensional arrays, Strings, Vectors, Wrapper classes.	13
3	Inheritance - Inheritance Basics, Using 'super', Creating Multilevel hierarchy, Method Overriding, Using Abstract Classes, Using final with Inheritance. Packages & Interfaces - Packages, Access protection in packages, Importing Packages, Interfaces. Exception Handling - Exception Handling Fundamentals – Exception Types, Uncaught Exceptions, Using try and catch, Multiple catch clauses, finally, Java's builtin Exceptions	13

4	Multithreaded Programming- Introduction, Creating threads, Extending the thread class, stopping & blocking thread, Life cycle of a thread, Using thread methods, Implementing the runnable interface. Event and GUI programming: The Applet Class, Types of Applets, Applet Basics, Applet Architecture, An Applet Skeleton, Simple Applet Display Methods, Requesting Repaint, The HTML APPLET tag. Event Handling - The delegation event model, Event Classes ActionEvent, KeyEvent&MouseEventClasses,EventListenerInterfaces– ActionListener, KeyListener & MouseListener interfaces. Using the Delegation Event Model. Window Fundamentals, Working with Frame Windows, Creating a Frame Window in an Applet. Creating a Windowed Program, Displaying information within a window. Introducing swing – two key swing features, components and containers, the swing packages, a simple swing application, event handling. Exploring Swing- JLabel, JTextField, JButton, Checkboxes , 13 Radio buttons , Jlist , JComboBox.	13
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TextBooks:

1. E Balagurusamy, Programming with Java – A Primer, Fourth Edition, Tata McGraw Hill Education Private Limited.
2. Herbert Schildt, Java : The Complete Reference, Seventh Edition, McGraw Hill Publication.

Reference Books:

1. HerbertSchildt,Java2-TheCompleteReference,FifthEdition,McGrawHill publication.
2. CayS.Horstmann,CoreJavaVolumeI–Fundamentals,PrenticeHall.
3. Somashekara,M.T.,Guru,D.S.,Manjunatha,K.S,ObjectOriented Programming with Java, EEE Edition, PHI.

**Ped
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Animations/ Role Plays/ Think-Pair-Share/ Predict-Observe- Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/Jigsaw/ Field based Learning/ Project Based Learning/ Mini Projects/ Hobby Projects/ Forum Theatre/ Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA	Semester	II
CourseTitle	ComputationalMathematics (Theory)		
CourseCode:	BCA-2.3	No.ofCredits	05
Contacthours	5Hours perweek	Durationof SEA/Exam	3 Hours
Formative AssessmentMarks	20	Summative AssessmentMarks	80

CourseOutcomes (COs):

Aftercompleting thiscoursesatisfactorily, astudentwillbeableto:

- Obtain an intuitive and working understanding of numerical methods for the basic problems of numerical analysis.
- Gainexperiencein theimplementationofnumericalmethodsusingacomputer.
- Traceerrorinthesemethodsandneedtoanalyseand predictit.
- Provide knowledge of various significant and fundamental concepts to inculcatein the students an adequate understanding of the application of Statistical Methods.
- Demonstratetheconceptsofnumericalmethodsusedfordifferentapplications

Unit	Description	Hours
1	Computer Arithmetic: Number System, Number representation, Floating point Arithmetic. Errors in numerical computation - Errors and their computation Solution of Algebraic and Transcendental equations: Introduction, the Bisection method, the method of False position, the Iterative method, Newton-Raphson method, Ramanujan's method. Interpolation: Introduction Finite differences- forward differences, backward differences, Central differences, Newton's formula for interpolation, Languages interpolation formula. Divideddifferences -Newton'sgeneralinterpolation formula	15
	LeastSquares -Introduction,leastsquarescurvefitting procedures-fittingastraightline,non-linearcurvefitting,	

2	curve fitting by sum of exponentials Numerical differentiation and integration - Numerical differentiation, Integration- Trapezoidal rule, Simpson's 1/3 rule and Simpson's 3/8 rule.	15
3	Matrices and linear system of equations: Basic definitions, matrix operations, transpose of a matrix, the inverse of a matrix, matrix norms. Solution of linear system: Direct methods- Matrix inversion method, Gaussian elimination method, Gauss-Jordan method, LU decomposition. Solution of linear systems- Iterative methods- Gauss-Seidel methods, Jacobi's method.	15
4	Numerical solution of ordinary differential equations: Solution by Taylor's series, Euler's method, Modified Euler's method, Runge-Kutta methods, Predictor-corrector methods - Adams-Moulton method, Milne's method, Boundary value problems- Finite difference method.	15
Text Book: 1. S.S. Sastry, Numerical Analysis, 3rd edition, PHI publication.		
Reference Books: 1. M.K. Jain, S.R.K. Iyengar & R.K. Jain, Numerical methods for Scientific and Engineering computation, 5th edition, New Age International publishers. 2. V. Rajaraman, Computer Oriented Numerical Methods, 3rd Edition, PHI, 2006		

Pedagogy: Lecture/ PPT/ Videos/ Animations/ Role Plays/ Think-Pair-Share/ Predict-Observe- Explain/ Demonstration/ Concept mapping/ Case Studies examples/ Tutorial/ Activity/ Flipped Classroom/ Jigsaw/ Field based Learning/ Project Based Learning/

Mini Projects/ Hobby Projects/ Forum Theatre/ Dance/ Problem Based Learning/ Game Based Learning/ Group Discussion/ Collaborative Learning/ Experiential Learning / Self Directed Learning etc.

ProgramName	BCA	Semester	II
CourseTitle	DataStructuresLab		
CourseCode:	BCA-2.4	No.ofCredits	02
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative AssessmentMarks	10	Summative AssessmentMarks	40

PART-A

1. Program to sort the given list using selection sort technique.
2. Program to sort the given list using insertion sort technique
3. Program to solve Tower of Hanoi using Recursion
4. Program to reverse string using Stack
5. Program to search an element using recursive binary search technique.
6. Program to implement Stack operations using arrays.
7. Program to implement Queue operations using arrays.
8. Program to implement dynamic array. Find smallest and largest element.

PART-B

1. Program to sort the given list using merge sort technique.
2. Program to implement circular queue using array.
3. Program to sort the given list using quick sort technique.
4. Program to implement Stack operations using linked list.
5. Program to implement Queue operations using linked list.
6. Program to evaluate postfix expression.
7. Program to perform insert node at the end, delete a given node and display contents of single linked list.
8. Menu driven program for the following operations on Binary Search Tree (BST) of Integers

- (a) Create a BST of N Integers
- (b) Traverse the BST in Inorder, Preorder and PostOrder.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A Writing: 7 Marks Execution: 8 Marks	15 Marks
Program-2	PART-B Writing: 10 Marks Execution: 10 Marks	20 Marks
Practical Record		05 Marks
Total		40 Marks

ProgramName	BCA	Semester	II
CourseTitle	ObjectOriented ProgrammingLab		
CourseCode:	BCA-2.5	No.ofCredits	02
Contacthours	4Hours perweek	Durationof SEA/Exam	3 Hours
Formative Assessment Marks	10	Summative Assessment Marks	40

PART-A

1. Program to accept student name and marks in three subjects. Find the total marks, average and grade (depending on the average marks).
2. Program, which reads two numbers having same number of digits. The program outputs the sum of product of corresponding digits.(Hint Input 327 and 539 output $3 \times 5 + 2 \times 3 + 7 \times 9 = 84$)
3. Program to input Start and End limits and print all Fibonacci numbers between the ranges.(Use for loop)
4. Define a class named Pay with data members String name, double salary, double da, double hra, double pf, double grossSal, double netSal and methods: Pay(String n, double s) - Parameterized constructor to initialize the data members, void calculate() - to calculate the following salary components, and void display() - to display the employee name, salary and all salary components.

DearnessAllowance=15%of salary

HouseRentAllowance=10%ofsalary Provident

Fund = 12% of salary

GrossSalary=Salary+DearnessAllowance+HouseRentAllowance

$\text{NetSalary} = \text{GrossSalary} - \text{ProvidentFund}$

Write a main method to create object of the class and call the methods to compute and display the salary details. [class basics]

5. Program to create a class DISTANCE with the data members feet and inches. Use a constructor to read the data and a member function Sum () to add two distances by using objects as method arguments and show the result. (Input and output of inches should be less than 12.).
6. Program to create a class "Matrix" that would contain integer values having varied numbers of columns for each row. Print row-wise sum.
7. Program to extract portion of character string and print extracted string. Assume that 'n' characters extracted starting from mth character position.
8. Program to add, remove and display elements of a Vector.

PART-B

1. Create a class named 'Member' having data members: Name, Age, PhoneNumber, Place and Salary. It also has a method named 'printSalary' which prints the salary of the members. Two classes 'Employee' and 'Manager' inherit the 'Member' class. The 'Employee' and 'Manager' classes have data members 'specialization' and 'department' respectively. Now, assign name, age, phone number, address and salary to an employee and a manager by making an object of both of these classes and print the same. [inheritance]
2. Program to implement the following class hierarchy:
Student: id, name
StudentExam (derived from Student): Marks of 3 subjects, total marks
StudentResult (derived from StudentExam) : percentage, grade
Define appropriate methods to accept and calculate grade based on existing criteria and display details of N students
3. Write a Program to calculate marks of a student using multiple inheritance implemented through interface. Class Student with data members rollNo, name, String cls and methods to set and put data.

Create another class test extended by class Student with data members mark1, mark2, mark3 and methods to set and put data.

Create interface sports with members sportsWt = 5 and putWt().

Now let the class results extend class test and implement interface sports. Write a Java program to read required data and display details in a neat format.

4. Write a Program to create an abstract class named shape that contains two integers and an empty method named print Area().
Provide three classes named Rectangle, Triangle and Ellipse such that each one of the classes extends the class shape. Each one of the class contains only the method print Area() that print the area of the given shape. [Abstract class].
5. Create a package to convert temperature in centigrade into Fahrenheit, and one more package to calculate the simple Interest. Implement both package in the Main () by accepting the required inputs for each application.
6. Write a Program that implements a multi-threaded program has three threads. First thread generates a random integer every second, and if the value is even, second thread computes the square of the number and prints. If the value is odd the third thread will print the value of cube of the number. [Multithreading]
7. Program that creates a user interface to perform basic integer operations.
The user enters two numbers in the TextFields - Num1 and Num2. The result of operations must be displayed in the Result TextField when the “=” button is clicked. Appropriate Exception handling message to be displayed in the Result TextField when Num1 or Num2 is not an integer or Num2 is Zero when division operation is applied.
8. Using the swing components, design the frame for shopping a book that accepts book code, book name, and Price. Calculate the discount on code as follows.

Code	Discount rate
101	15%
102	20%
103	25%
Any other	5%

Find the discount amount and Net bill amount. Display the bill.

Evaluation Scheme for Lab Examination:

Assessment Criteria		
Program-1	PART-A	15 Marks
	Writing: 7 Marks Execution: 8 Marks	

Program-2	PART-B Writing:10MarksExecution:10Marks	20 Marks
Practical Record		05Marks
Total		40 Marks

QuestionsPaperforPatternCoreSubjects

Duration:3Hours

Max.Marks:80

Note:AnsweranytenQuestionsfromPart-A.AndonefullQuestionsfromeachunitin Part-B

Part-A

1.

10*2=20

- a.
- b.
- c.
- d.
- e.
- f.
- g.
- h.
- i.
- j.
- k.
- l.

Part-B

UNIT-I,II,III,IV

Eachunitcontainstwomainquestionsanditcarry15Marks. Each main questions contain 3 or more sub question.

4*15=60

UNIT-I

2.

- a.
- b.
- c.

3.

- a.
- b.
- c.